Improved A / LegalSpan

Task and source

The Phase 1 task is 128-word legal continuation given a 512-word prompt. The four official sources are Acts, High Court, Supreme Court, and Tribunal material. This report is a later reconstruction based on recovered source and archived run evidence, not an executed training notebook.

Architecture

The dense 12-block causal backbone gains a 128-dimensional prompt route in its upper four blocks. Role and continuity cues bias access to prompt states; a learned gate mixes vocabulary generation with prompt copying. The route respects causal boundaries.

Stage	512-word prompt -> Own byte BPE -> 12 causal blocks
Continuation	4 routed upper blocks -> Copy / vocabulary gate -> 128-word continuation

Token and optimization science

The model learns by minimizing next-token cross-entropy under causal masking. An attention head compares query/key vectors, obtains a weighted value mixture, and updates the residual stream; rotary phases give position-sensitive comparisons. Decoding then turns logits into a 128-word continuation. Tokenizer choice affects sequence length, vocabulary utilization, and the optimization budget. The original source config and scripts identify the actual optimizer, schedule, context and generation policy.

Reported outcome

0.8341 (historical Codabench screenshot). Historical screenshot numbers are not freshly retrieved leaderboard results. Archived local metric files, when present, provide separately measured BERTScore, ROUGE-L and BLEU-4; no metric is invented for runs without such evidence.

Reproduction boundary

The code package contains the recovered model, tokenizer, training, evaluation and export scripts, plus original small logs/metrics. The complete ZIP holds selected checkpoint weights and official data, which are too large or inappropriate to duplicate in GitHub. Re-execution requires their original hashes and path remapping.

Configured dimensions

The following values are read directly from the original configuration. A continued-training config may inherit the parent model graph; that relationship is identified in the README and source code.

Setting	Archived value
vocab_size	32768
max_seq_len	8192
n_layers	12
d_model	768
n_heads	12
d_head	64
d_ff	2048
rope_theta	10000.0
dropout	0.0
gradient_checkpointing	True
route_layers	4
route_dim	128
route_mode	full

Archived evaluations

This table is a small sample of per-policy and per-checkpoint records, not a single comparable leaderboard ranking. Full machine-readable values and names are in evidence/archived_metric_index.json.

Metric file	BERTScore F1	ROUGE-L F1	BLEU-4
control/confirmation.metrics.json	0.8651	0.3476	0.2635
control/final_policy0_dev.metrics.json	0.8525	0.3617	0.2645
control/final_policy1_dev.metrics.json	0.8543	0.3637	0.2665
control/final_policy2_dev.metrics.json	0.8652	0.3474	0.2620
control/targets_104957342_policy0_dev.metrics...	0.8429	0.3294	0.2259
control/targets_104957342_policy1_dev.metrics...	0.8461	0.3306	0.2289
control/targets_104957342_policy2_dev.metrics...	0.8571	0.3150	0.2119
generic/confirmation.metrics.json	0.8653	0.3492	0.2635

Evidence count and caveat

Copied 54 compact log/metric/manifest artifacts from CS781_Improved_A_LegalSpan_Complete_20261006.zip. They support the run lineage and measured local evaluations. A lower generation loss need not imply better BERTScore or legal accuracy; checkpoint and decoding policies must be evaluated separately. Original October 6 source archive contains no .ipynb file, so a

later illustrated notebook is explicitly marked as reconstructed.